



Yunqiang Pei (Simon)

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Research Interest

Augmented Reality × AI, Human-Agent Alignment, Physiological Computing

Professional Experience

Mar. 2025 – Mar. 2026

KAIST, *South Korean*

- Visiting student researcher conducting research in the WIT Lab, under the supervision of Professor Ian Oakley

Education

Sep. 2022 – Present

Ph.D. Candidate in Electronic Information, *University of Electronic Science and Technology of China (UESTC)*, Chengdu, China

- **GPA:** 3.84 / 4.0
- School of Computer Science and Engineering

Sep. 2018 – Jun. 2021

M.Sc. in Computer Science and Technology, *Southwest University of Science and Technology*, Mianyang, China

- **GPA:** 90.88 / 100.0 (**Ranked 1st** in the whole grade)
- School of Computer Science and Technology

Sep. 2014 – Jun. 2018

B.S. in Internet of Things Engineering, *Sichuan University of Science and Engineering*, Zigong, China

- **GPA:** 3.7 / 5.0 (**Ranked 1st** in Major)
- School of Computer Science

Employment

Jul. 2021 – Nov. 2021

Software engineer, Thoughtworks, *Chengdu, Sichuan, China*.

Honors and Awards

Class of 2026

Outstanding Graduate Student (Projected Honor)

2025

Academic Rising Star Honor of UESTC

2024

🏆 **Best Paper Nomination**, ACM Multimedia (MM) 2024

2024

First-Class Scholarship

2023

Second Prize, National University Student Five Minute Research Presentation (English)

2023, 2025

Second-Class Scholarship (Awarded twice)

2022

Freshman Scholarship

2021

Outstanding Master's Thesis

2019

🏆 **Best Paper Honorable Mention Award**, ICVRV 2019

2015 – 2017

National Scholarship (Awarded for **two consecutive years**)

2018

Sichuan Province Outstanding Graduate

2014 – 2017

First-Class Scholarship (Awarded for three consecutive years)

Publications

Quick summary since 2019: First Author(8), Awarded Paper(1), Best Paper Nomination (1), MM(3), VR (poster × 2), UIST(1), AAAI(1), ICVRV(1), PRAI(1), SCIENCE CHINA Information Sciences(1), Journal of System Simulation(1), Journal of Graphics(2)

- C1 **Yunqiang Pei**, Hongrong Yang, Kaiyue Zhang, Guoqing Wang*, Peng Wang, Chaoning Zhang, Yang Yang, Heng Tao Shen. InteractGuide: LLM-Enhanced Multimodal Reasoning for User-Centric Interaction Recommendations in AR-HRI Authoring. *ACM Multimedia* (MM Oral Paper). 2025
- C2 **Yunqiang Pei**, Renming Huang, Mingfeng Zha, Guoqing Wang*, Peng Wang, Kang Qiao, Yang Yang, Heng Tao Shen. AttentionAR: AR Adaptation and Warning for Real-World Safety via Attention Modeling and MLLM Reasoning. *ACM Symposium on User Interface Software and Technology* (UIST Paper). 2025
- C3 **Yunqiang Pei**, Kaiyue Zhang, Hongrong Yang, Yong Tao, Qihang Tang, Jialei Tang, Guoqing Wang*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Improving Interaction Comfort in Authoring Tasks in AR-HRI through Dynamic Dual-Layer Interaction Adjustment. *ACM Multimedia* (🏆 **Best Paper Nomination**). 2024
- C4 **Yunqiang Pei**, Qihang Tang, Jialei Tang, Mingfeng Zha, Dongyu Xie, Guoqing Wang*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Emotion Recognition in HMDs: A Multi-task Approach Using Physiological Signals and Occluded Faces. *ACM Multimedia* (MM Paper). 2024
- C5 **Yunqiang Pei**, Kaiyue Zhang, Ziyang Lu, Mingfeng Zha, Guoqing Wang*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Toward Optimized AR-based Human-Robot Interaction Ergonomics: Modeling and Predicting Interaction Comfort. *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR Poster). 2024
- C6 Ziyang Lu, **Yunqiang Pei**, Guoqing Wang*, Peiwei Li, Yang Yang, Yinjie Lei, Hengtao Shen. ScanERU: Interactive 3D Visual Grounding Based on Embodied Reference Understanding. *The 38th Annual AAAI Conference on Artificial Intelligence*. 2024
- C7 **Yunqiang Pei**, Renming Huang, Guoqing Wang*, Yang Yang, Heng Tao Shen. Multimodal Apology: Using WebXR to Repair Trust with Virtual Companion. *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR Poster). 2023
- C8 Shanshan Xiang, Wensheng Tang, **Yunqiang Pei**, Yadong Wu*, Yonghui Chen, Fupan Wang, Zhiwei Hou. IV-WA: An Information Visual Tool with WebAR. *International Conference on Pattern Recognition and Artificial Intelligence* (PRAI). 2021
- C9 **Yunqiang Pei**, Yadong Wu*, Song Wang, Fupan Wang, Hongyu Jiang, Shijian Xu, Jinqian Zhou. Wa vis: A Web-based Augmented Reality Text Data Visual Analysis Tool. *International conference on virtual reality and visualization* (ICVRV, 🏆 **Best Paper Honorable Mention Award** ★). 2019
- J1 Guoqing Wang (Advisor)*, **Yunqiang Pei**, Yang Yang, Xing Xu, Zheng Wang, Hengtao Shen. A Review and Outlook of Shared Multi-Modal Trustworthy Human-Robot Interaction Research, *SCIENCE CHINA Information Sciences*. 2023
- J2 Chunhong Liu, Song Wang*, Fupan Wang, Wensheng Tang, **Yunqiang Pei**, Dongsheng Tian, Yadong Wu. AR-assisted Sign Language Letter Recognition Method Based on Improved MobileNet Network, *Journal of System Simulation*. 2023
- J3 Nanshan Liu, **Yunqiang Pei**, Hao Jiang, Yongguo Han, Yadong Wu*, Fupan Wang, Siheng Yi. WebAR garbage classification information visualization method based on VD-MobileNet network, *Journal of Graphics*. 2022
- J4 **Yunqiang Pei**, Yadong Wu*, Fupan Wang, Xiaorong Zhang, Hongyu Jiang, Shijian Xu, Wensheng Tang. IV LKWA: an information visual analysis tool with advanced L-K optical flow based WebAR, *Journal of Graphics*. 2020

Peer Review Service

AAAI, UIST, CHI, CSCW, ISMAR, VRST, UbiComp/ISWC, IUI, MM, DIS, IDC, IMX

Winter and Summer Vacation Study Program

Jan. 2023 – Feb. 2023

I participated in the Machine Learning Winter Programme organised by **Girton College, Cambridge** (Online)

Conference Volunteering Experience

Jul. 2019

Student Volunteer at **ChinaVIS 2019**, Chengdu, China.

Apr. 2025

Student Volunteer at **ACM CHI 2025**, Yokohama, Japan.

Oct. 2025

Student Volunteer at **ISMAR 2025**, Daejeon, South Korea.

Hardware Experience (Software Utilization)

HMD & Glass

Hololens 2, XReal Air 2 Ultra, 1S, Beam Pro, Eye

PPG & GSR

Shimmer3R GSR+

EEG & IMU

Brainco Oxyzen, OpenBCI Ganglion

Invited Talks

Mar. 2025

Enhancing XR Interactions: Comfort, Emotion, and Safety — Lunch Seminar, WIT Lab, KAIST

Selected Press Coverage

Nov. 2025

UESTC Takes Me to See the World · Exploring the Future of Human-AI Symbiosis at the Intersection of AR and AI

Languages and Levels

Chinese

Proficient

English

Advanced

Korean

Intermediate